STOCK 1 I-BEAM · REV C · FINAL DESIGN

EMCH 308 Team 6 · AA I 3 × 2.03, 6061-T6 · 8.40 in long · 3-point bend, 8.00 in span

Full-width flange pads under every roller, tips notched elsewhere · inches from mid-span (CL) · Abaqus job Stock1_RevC_v2, Nov 8, 2026 (0 warnings)

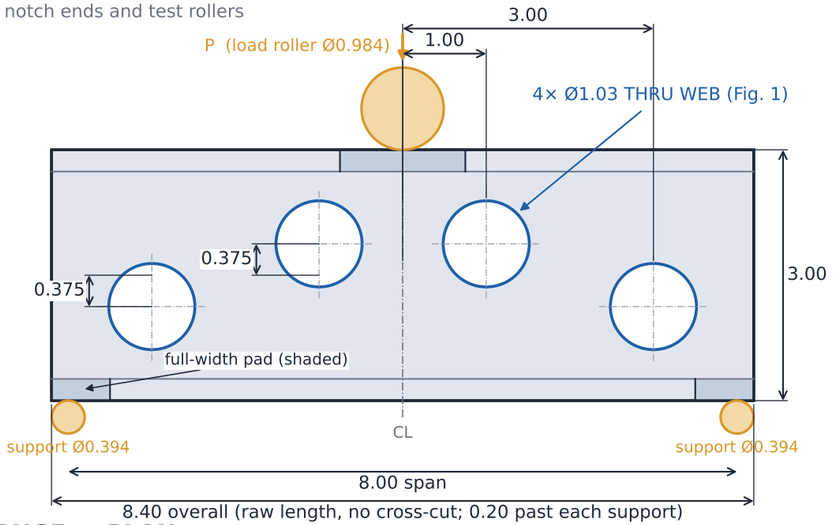
WEIGHT PASS
0.914 lb
limit ≤ 1.00 lb · margin 0.086 lb

FACTOR OF SAFETY PASS
1.770
required ≥ 1.25 · Abaqus offset method

COST PASS
$131.78
limit ≤ $600 · course cost sheet

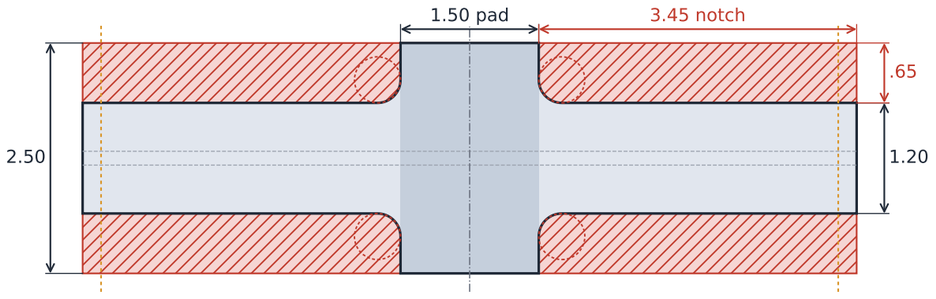
SIDE ELEVATION

web holes, notch ends and test rollers



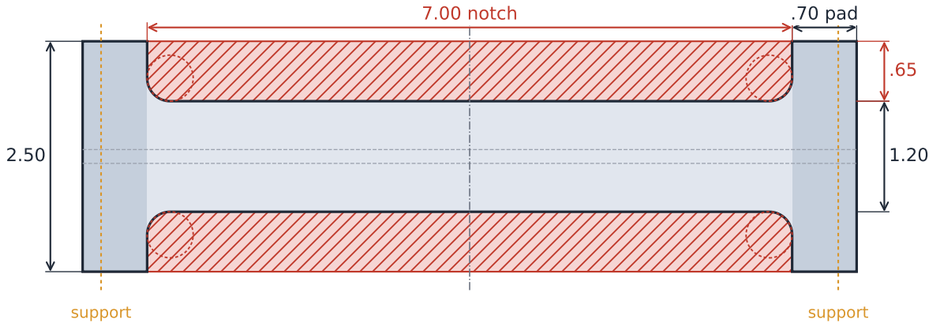
TOP FLANGE · PLAN

load-roller pad 1.50 long at CL · 4 notches 0.65 deep, 3.45 long, run out the ends · 4 ×0.50 relief drills (R0.25 corners)



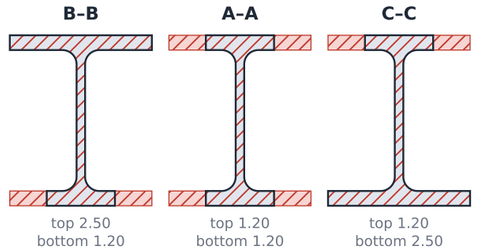
BOTTOM FLANGE · PLAN

support pads 0.70 long at both ends · 2 notches 0.65 deep, 7.00 long · 4 ×0.50 relief drills (R0.25 corners)



SECTIONS

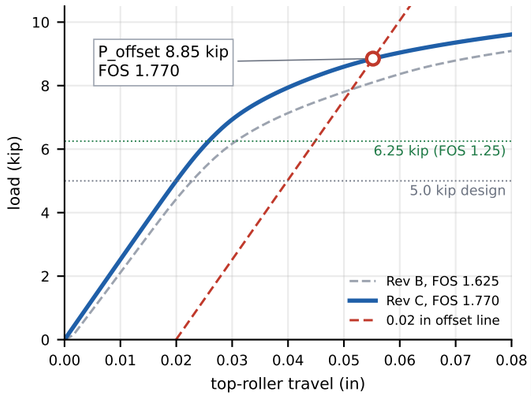
B–B at CL · A–A between pads · C–C over a support



flange 0.26 · web 0.15 · R0.25 fillets · height 3.00

FOS · ABAQUS LOAD–DISPLACEMENT

0.02 in offset · C3D10, 234,665 nodes · 105 increments



REQUIREMENTS CHECK

Requirement	Limit	Rev C	
Weight	≤ 1.00 lb	0.914 lb (mesh = CAD to 0.003 %)	PASS
Factor of safety	≥ 1.25	1.770 (P_offset 8,850 lbf)	PASS
Cost	≤ $600	$131.78	PASS
Span / height	fixed	8.00 / 3.00 in	PASS
Fig. 1 holes	4 × Ø1 ±.02	4 × Ø1.03 (team choice, +.01 over)	NOTE

At 6,250 lbf: curve 3.6 % below linear, est. permanent set 0.0009 in (offset allows 0.020) · linear to about 5,411 lbf

COST · course cost sheet

Material, Stock 1 (8.40 in raw)	$55.02
Saw: set-up + 14 cuts	$40.05
Drill: set-up + 12 holes	$36.71
Total	**$131.78**

(0.1667 + 0.0167·n) h × $100/h · material 8.4 × 2.03/12 × $38.72
with setups the sheet omits: $183.46 max

WEIGHT · 6061, 0.098 lb/in³

Stock 1 blank, 8.40 in	1.421
4 Ø 1.03 web holes	-0.049
4 top-flange notches	-0.227
2 bottom-flange notches	-0.231
Finished beam (lb)	**0.914**

CAD 9.3252 in³; Abaqus mesh agrees to 0.003 %
margin 0.086 lb (0.928 lb if notches 0.02 shallow)

FEATURE SCHEDULE

Tag	Size	Qty	Location (from CL, web centreline)	
A	Ø1.03 (1-1/32)	4	web, z ±1.00 (y +0.375), ±3.00 (y −0.375)	drill
B	Ø0.50 relief	8	top z ±1.00, bottom z ±3.25; x ±0.85	drill first
C	notch 0.65 deep	4	top flange, |z| 0.75 → end, both tips, R0.25	saw
D	notch 0.65 deep	2	bottom flange, z −3.50 → +3.50, both tips, R0.25	saw

TITLE	REV	MATERIAL
Stock 1 I-Beam, Rev C, Final Design	C	6061-T6 Al
PROJECT	**DATE**	**UNITS / TOL**
EMCH 308 · USC Team 6 / MTC	2026-11-08	in · ±0.02

Notes: drill relief holes before any saw cut · saw tangent to the relief holes (no sharp inside corners) · cut the flange only, never the web or fillets · break edges 0.02 max

Manufacturing drawing: USC06_MTC06_IBeam_RevC (3 sheets) · model: abaqus/python/final/Stock1_RevC_v1.py · FoS: FOS_Stock1_RevC_v3.py (job Stock1_RevC_v2)